Basic Statistical Concepts
Part I

Overview

- Levels of measurement
- Standardized and unstandardized variables
- Bivariate correlation and regression

Levels of Measurement

Psychological measurement is about differences.

To study differences, we must understand:

- Variability
- Distribution
- Central tendency
- Association between variables

Without variability, measurement would be meaningless.

Levels of Measurement

Variables can take on many different forms and levels of sophistication

The relationship between *what is being measured* and *the numbers that represent what is being measured* is known as the level of measurement

Variables can be categorical or continuous, and can have different levels of measurement

Levels of Measurement

A categorical variable is made up categories

A categorical variable that you should be familiar with already is your species:

human, domestic cat or fruit bat

Levels of Measurement

A categorical variable is one that names distinct entities

In its simplest form it names just two distinct types of things, for example male or female

This is known as a <u>binary variable</u>

- being alive or dead,

- pregnant or not

- responding “yes” or “no” to a question

In all cases there are just <u>two categories,</u> and entity can be placed into only one of the two categories

Levels of Measurement

When there are more than two possibilities, the variable is said to be a <u>nominal variable</u>

- Hair color: blonde, brown, red

- Political affiliation: republican, democrat, independent or other

Levels of Measurement

So far, the categorical variables have been unordered (e.g. different brands of Coke), but they can be ordered

When categories are ordered, the variable is known as an <u>ordinal variable</u>

Ordinal data tell us not only that things have occurred, but also the order in which they occurred

Levels of Measurement

A continuous variable is one that gives us a score for each person and can take on any value on the measurement scale that we are using

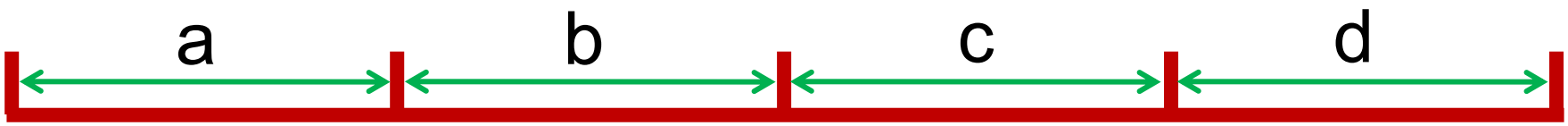
The first type of continuous variable that you might encounter is an interval variable

To say that data are interval, we must be certain that equal intervals on the scale represent equal differences in the property being measured

Levels of Measurement

For example, on "www.ratemyprofessors.com" students are encouraged to rate their lecturers on several dimensions (i.e., helpfulness, clarity, etc.)

Each dimension is evaluated a 5-point scale



$$a = b = c = d$$

Levels of Measurement

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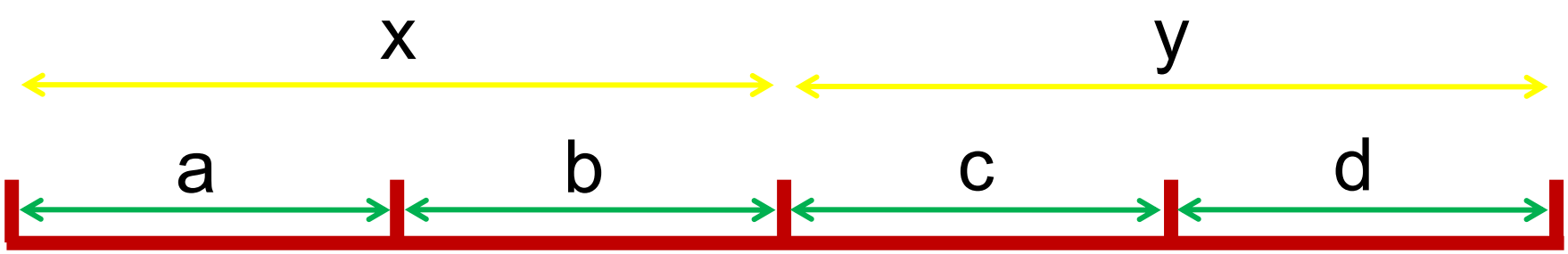


$$x = y$$

Similarly, difference in helpfulness between ratings of 1 and 3 should be identical to difference between ratings of 3 and 5

Levels of Measurement

Ratio variables go a step further than interval data by requiring that the ratios of values along the scale should be meaningful

For this to be true, the scale must have a true and meaningful zero point

Levels of Measurement

For example, data obtained from an IQ test are interval variable, because;

(1) If the score of a person from an IQ test is “0”, it doesn’t mean that he/she doesn’t have any intelligence

So this zero point isn’t meaningful

(2) If scores of Aslı and Suna from an IQ test are 60 and 120 respectively, we don’t conclude that Suna is twice as intelligent as Aslı

So the ratios of values along the scale are not meaningful

Levels of Measurement

The time to respond to something is a good example of a ratio variable

When we measure reaction time, it is true that the difference between 300 and 350 ms is the same as the difference between 210 and 260 ms

It is also true that distances along the scale are divisible:

A reaction time of 200ms is twice as long as a reaction time of 100 ms and twice as short as a reaction time of 400 ms

In this examle, there is a true and meaningful zero point

0 ms = no reaction

Levels of Measurement

Variables can be split into categorical and continuous, and within these types are different levels of measurement:

Categorical (entities are divided into distinct categories):

- Binary variable: There are only two categories

- Nominal variable: There more than two categories

- Ordinal variable: The same as a nominal variable but the categories have a logical order

Continuous (entities get a distinct score):

- Interval variable: Equal intervals on the variable represent equal differences in the property being measured

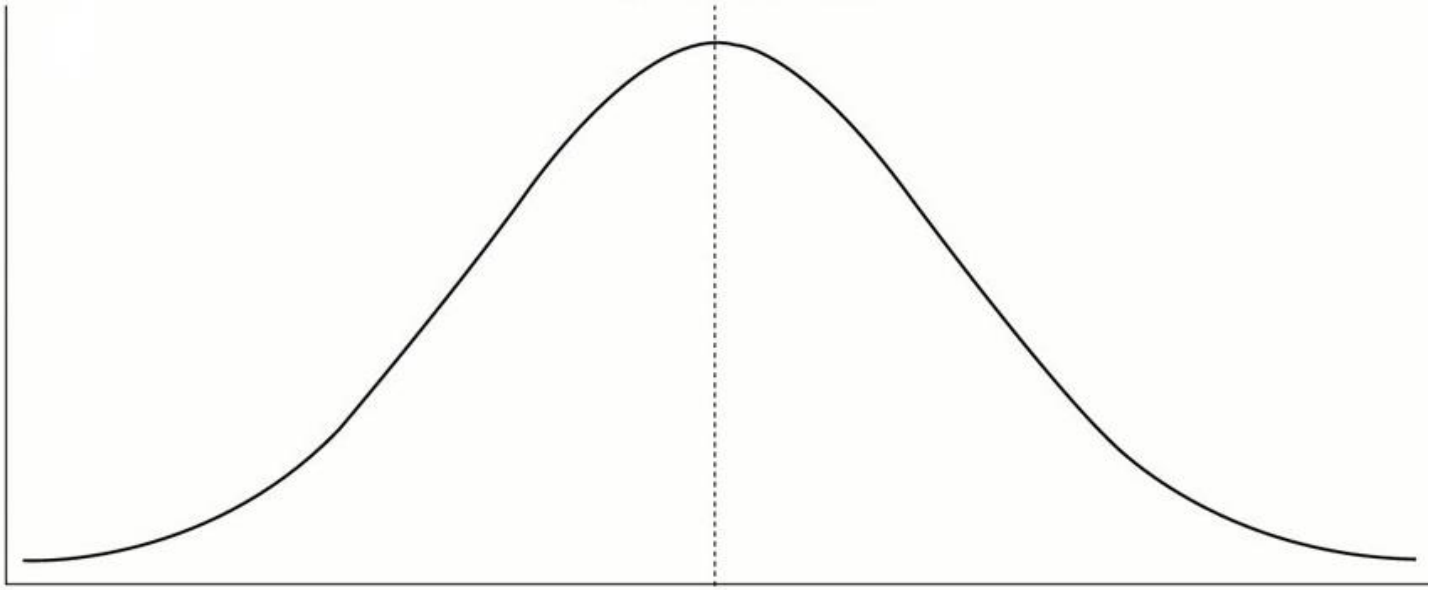
- Ratio variable: The same as an interval variable, but the ratios of scores on the scale must also make sense

Normal Distribution

Frequency distributions come in many different shapes and sizes

In ideal world our data would be distributed symmetrically around the center of all scores

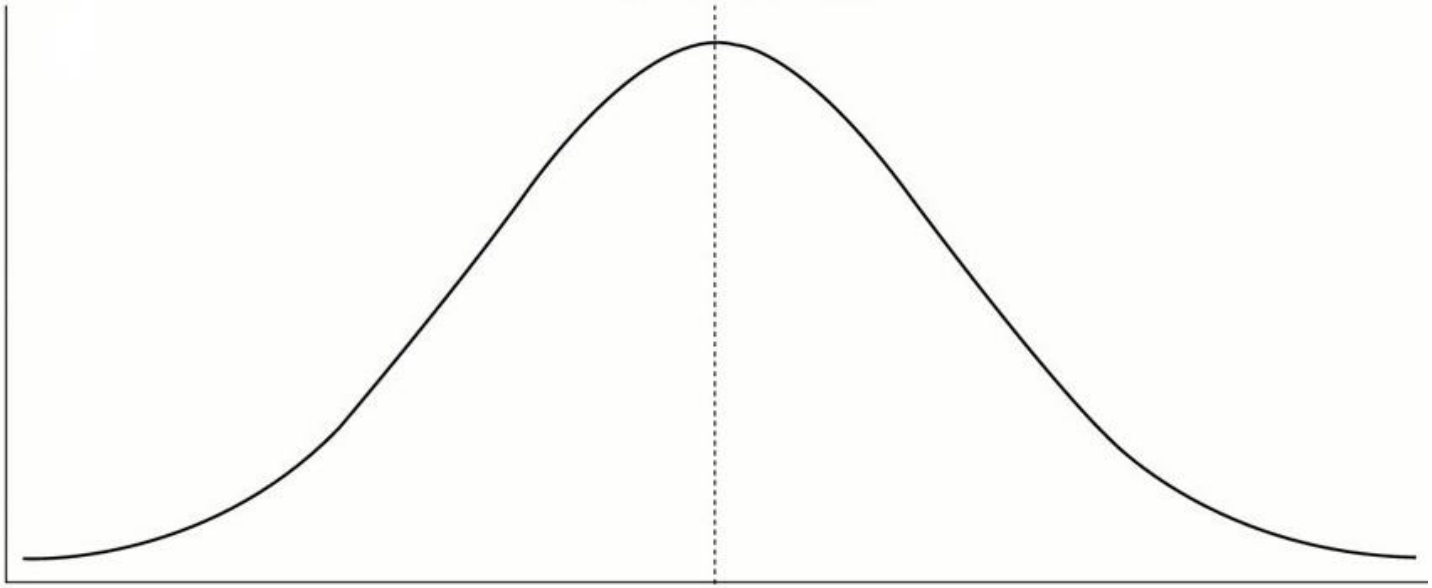
This is known as a <u>normal distribution</u> and is characterized by the bell-shape curve



Normal Distribution

There are three characteristics of a normal distribution:

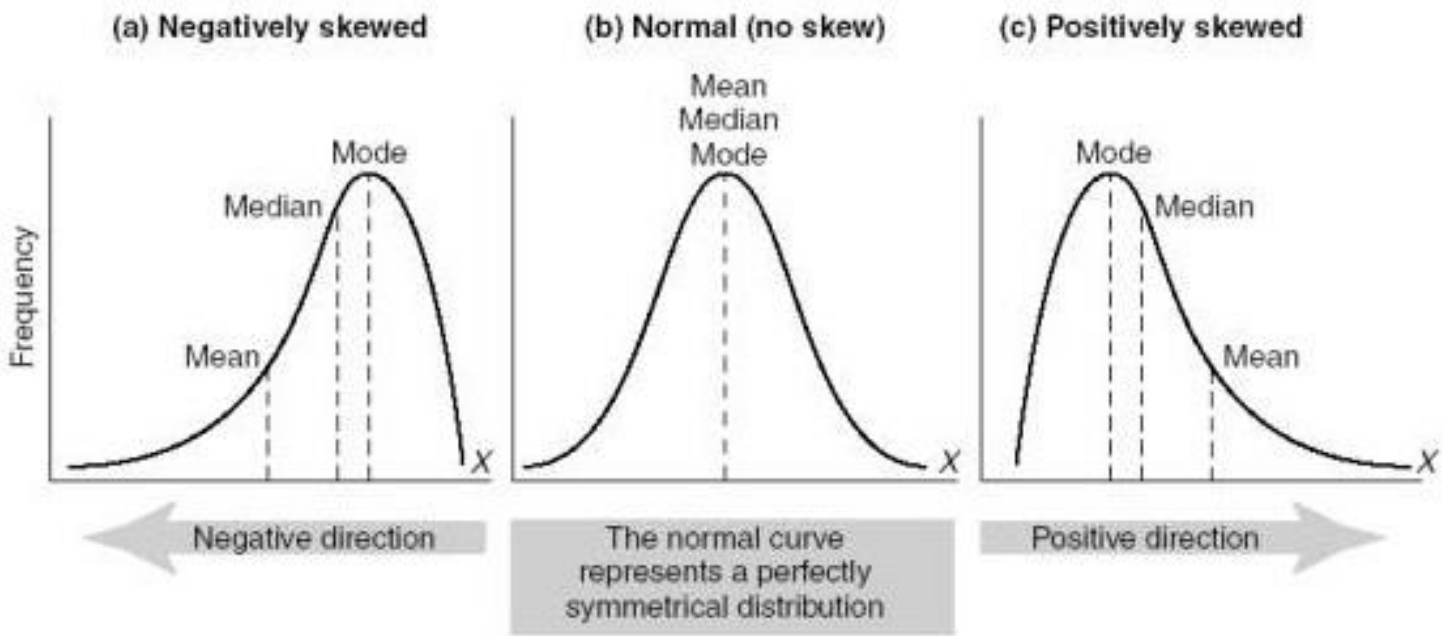
1. It is symmetrical about the mid-point of the horizontal axis
2. The values mean, median and mode are identical, and they are located in this mid-point
3. The “asymptotes” (tail ends) of the perfect curve never quite meet the horizontal axis



Normal Distribution

There are two main ways in which a distribution can deviate from normal:

1. Lack of symmetry (called skew)
2. Lack of pointyness (called kurtosis)

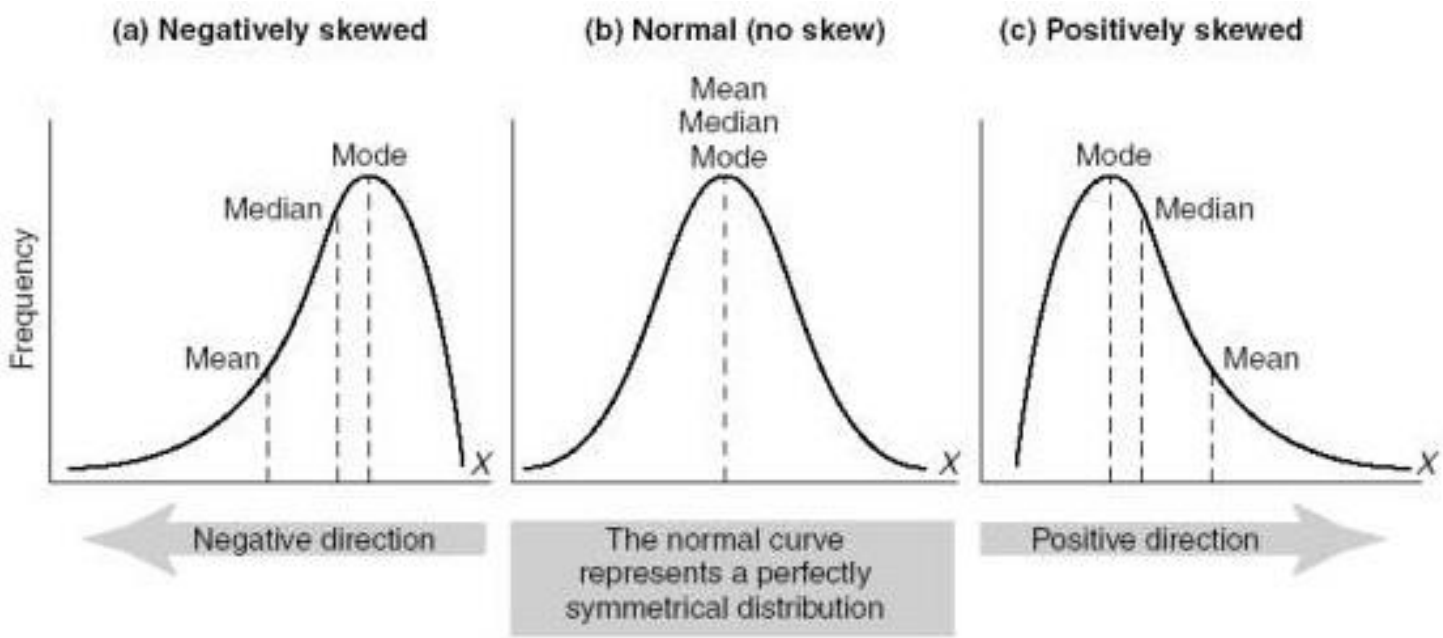


Normal Distribution

Skewed distributions are not symmetrical and instead the most frequent scores (the tall bars in the graph) are clustered at one end of the scale

A skewed distribution can be either:

Positively skewed or negatively skewed

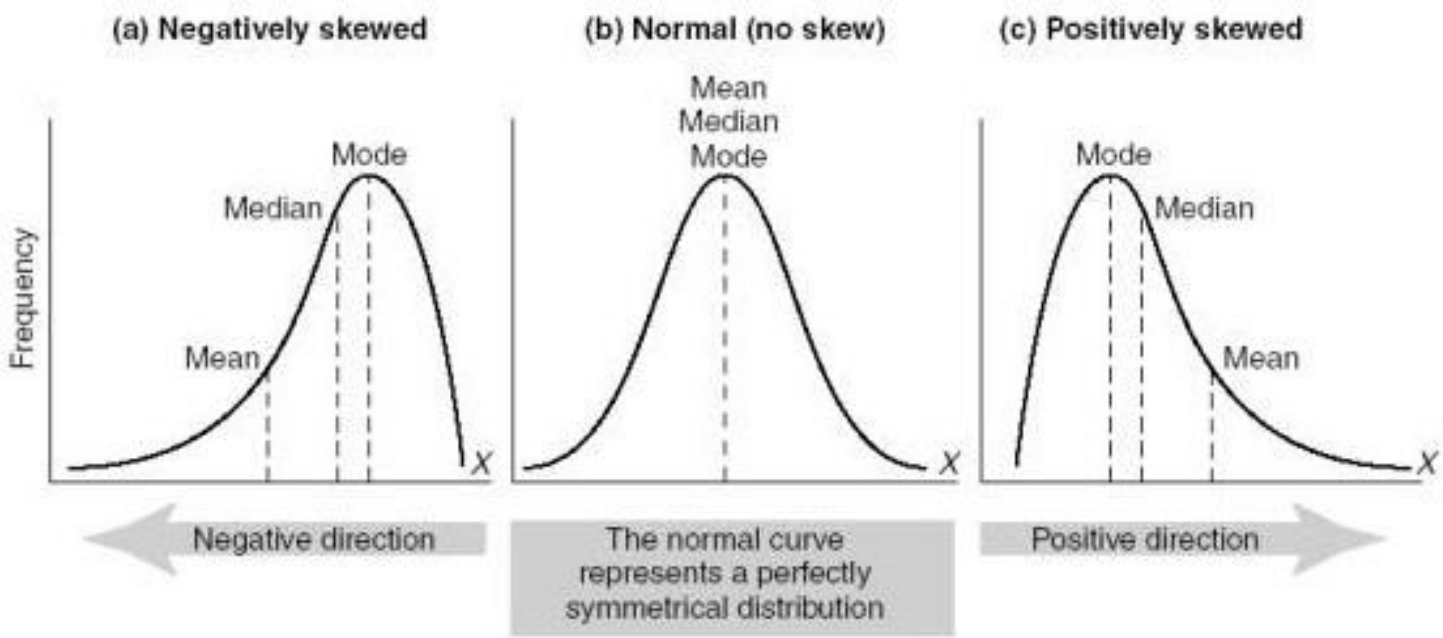


Normal Distribution

Positively skewed: The frequent scores are clustered at the lower scores and the tail points towards the higher scores

In a positively skewed distribution, more scores are to the right of the mode than to left

Therefore, in a positively skewed distribution the mean and median are bigger than the mode

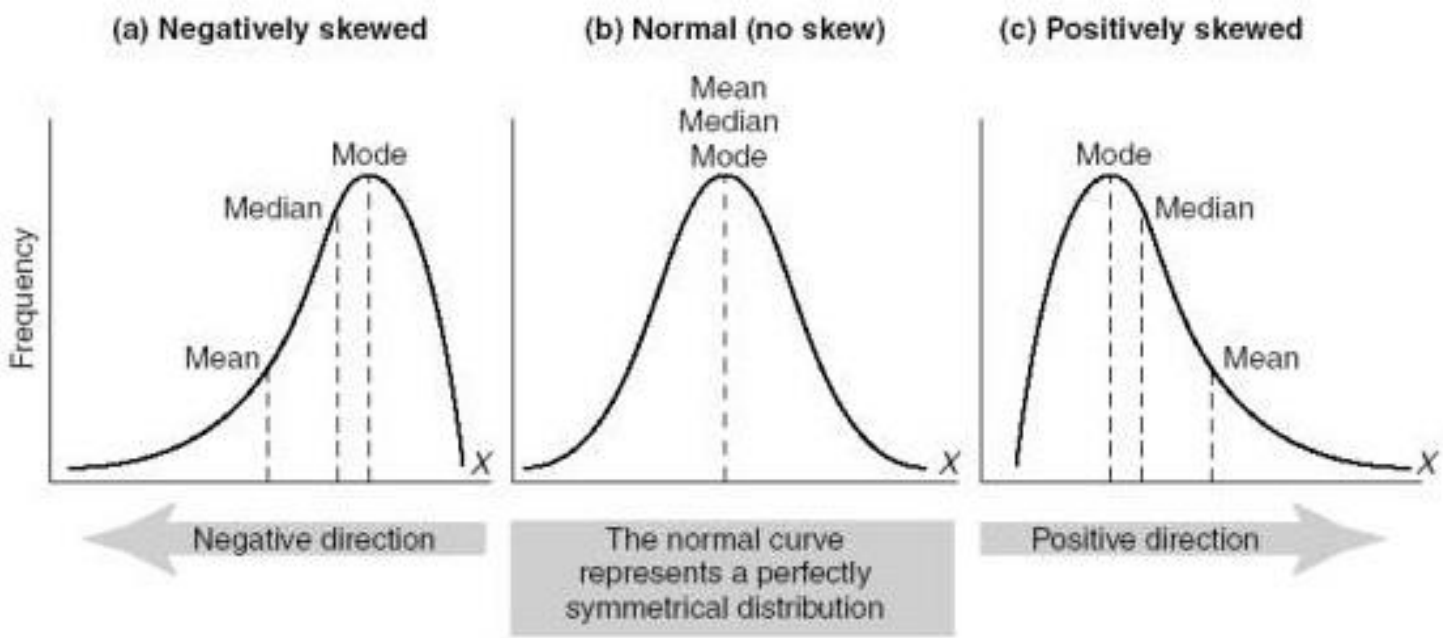


Normal Distribution

Negatively skewed: The frequent scores are clustered at the higher scores and the tail points towards the lower scores

In a negatively skewed distribution, more scores are to the left of the mode than to right

Therefore, in a negatively skewed distribution the mean and median are smaller than the mode



Standardized and unstandardized variables

- A *standardized variable* has a mean of 0 and a standard deviation of 1.00

- Converting raw scores to *z* scores is the most common way to standardize a variable:

$$z = \frac{X - M}{SD}$$

- Standardized estimates are calculated with standardized variables, and they are interpreted the same way for all variables

Standardized and unstandardized variables

- The range of absolute values for standardized estimates is sometimes, but not always, 0 to 1.00
- In contrast, the range of *unstandardized estimates* is determined by the original metrics (scales) of the unstandardized variables
- Unstandardized estimates cannot generally be directly compared across different variables
- The covariance is an example of an unstandardized statistic:

$$cov_{XY} = r_{XY}\, SD_X\, SD_Y$$